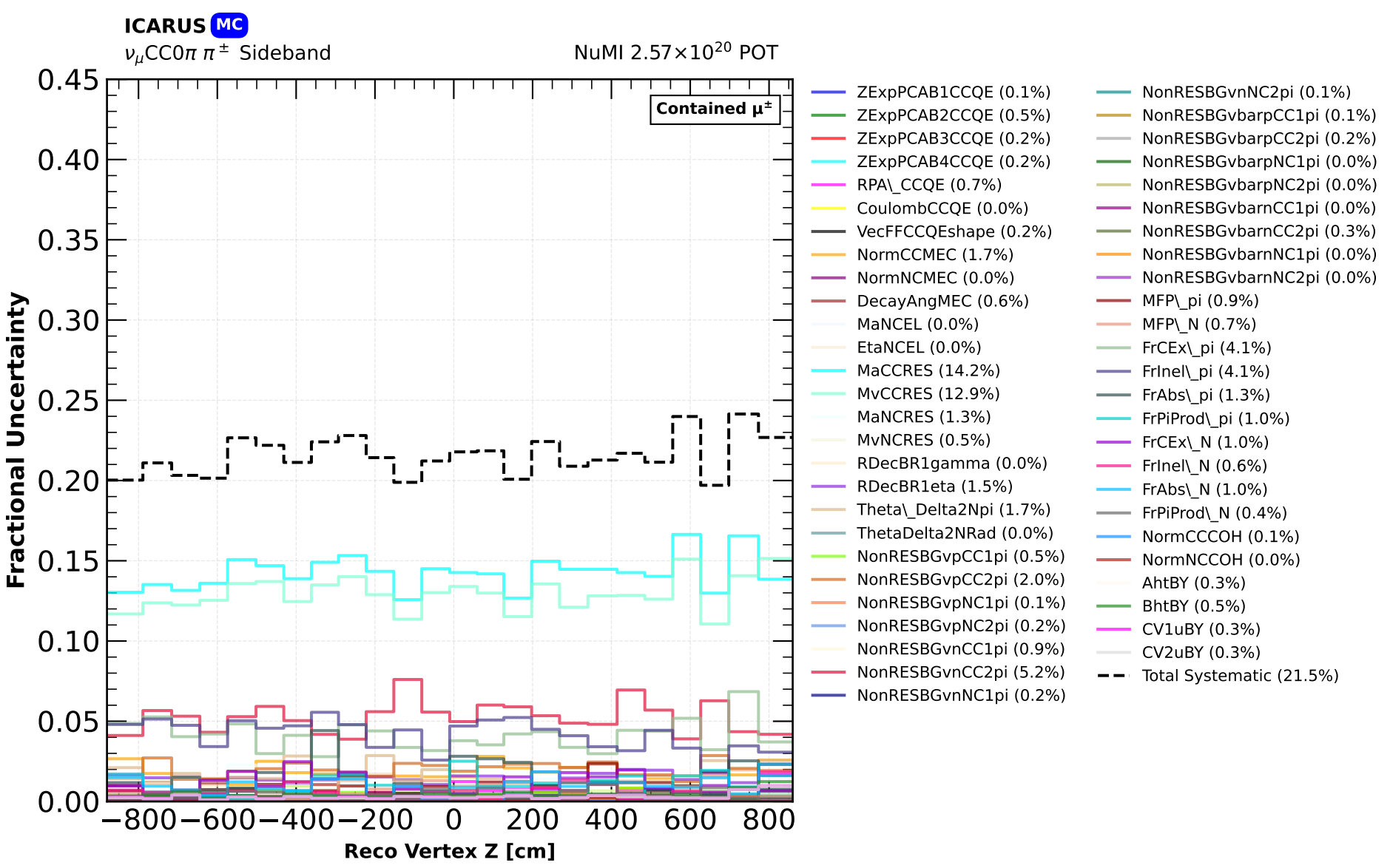
ICARUS MC
ν_μCC0π $\pi^\pm$ Sideband
NuMI 2.57×10^{20} POT
Contained μ±
Fractional Uncertainty
Reco Vertex Z [cm]
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
−800
−600
−400
−200
0
200
400
600
800
ZExpPCAB1CCQE (0.1%)
ZExpPCAB2CCQE (0.5%)
ZExpPCAB3CCQE (0.2%)
ZExpPCAB4CCQE (0.2%)
RPA_CCQE (0.7%)
CoulombCCQE (0.0%)
VecFFCCQEshape (0.2%)
NormCCMEC (1.7%)
NormNCMEC (0.0%)
DecayAngMEC (0.6%)
MaNCEL (0.0%)
EtaNCEL (0.0%)
MaCCRES (14.2%)
MvCCRES (12.9%)
MaNCRES (1.3%)
MvNCRES (0.5%)
RDecBR1gamma (0.0%)
RDecBR1eta (1.5%)
Theta_Delta2Npi (1.7%)
ThetaDelta2NRad (0.0%)
NonRESBGvpCC1pi (0.5%)
NonRESBGvpCC2pi (2.0%)
NonRESBGvpNC1pi (0.1%)
NonRESBGvpNC2pi (0.2%)
NonRESBGvnCC1pi (0.9%)
NonRESBGvnCC2pi (5.2%)
NonRESBGvnNC1pi (0.2%)
NonRESBGvnNC2pi (0.1%)
NonRESBGvbarpCC1pi (0.1%)
NonRESBGvbarpCC2pi (0.2%)
NonRESBGvbarpNC1pi (0.0%)
NonRESBGvbarpNC2pi (0.0%)
NonRESBGvbarnCC1pi (0.0%)
NonRESBGvbarnCC2pi (0.3%)
NonRESBGvbarnNC1pi (0.0%)
NonRESBGvbarnNC2pi (0.0%)
MFP_pi (0.9%)
MFP_N (0.7%)
FrCEx_pi (4.1%)
FrInel_pi (4.1%)
FrAbs_pi (1.3%)
FrPiProd_pi (1.0%)
FrCEx_N (1.0%)
FrInel_N (0.6%)
FrAbs_N (1.0%)
FrPiProd_N (0.4%)
NormCCCOH (0.1%)
NormNCCOH (0.0%)
AhtBY (0.3%)
BhtBY (0.5%)
CV1uBY (0.3%)
CV2uBY (0.3%)
Total Systematic (21.5%)